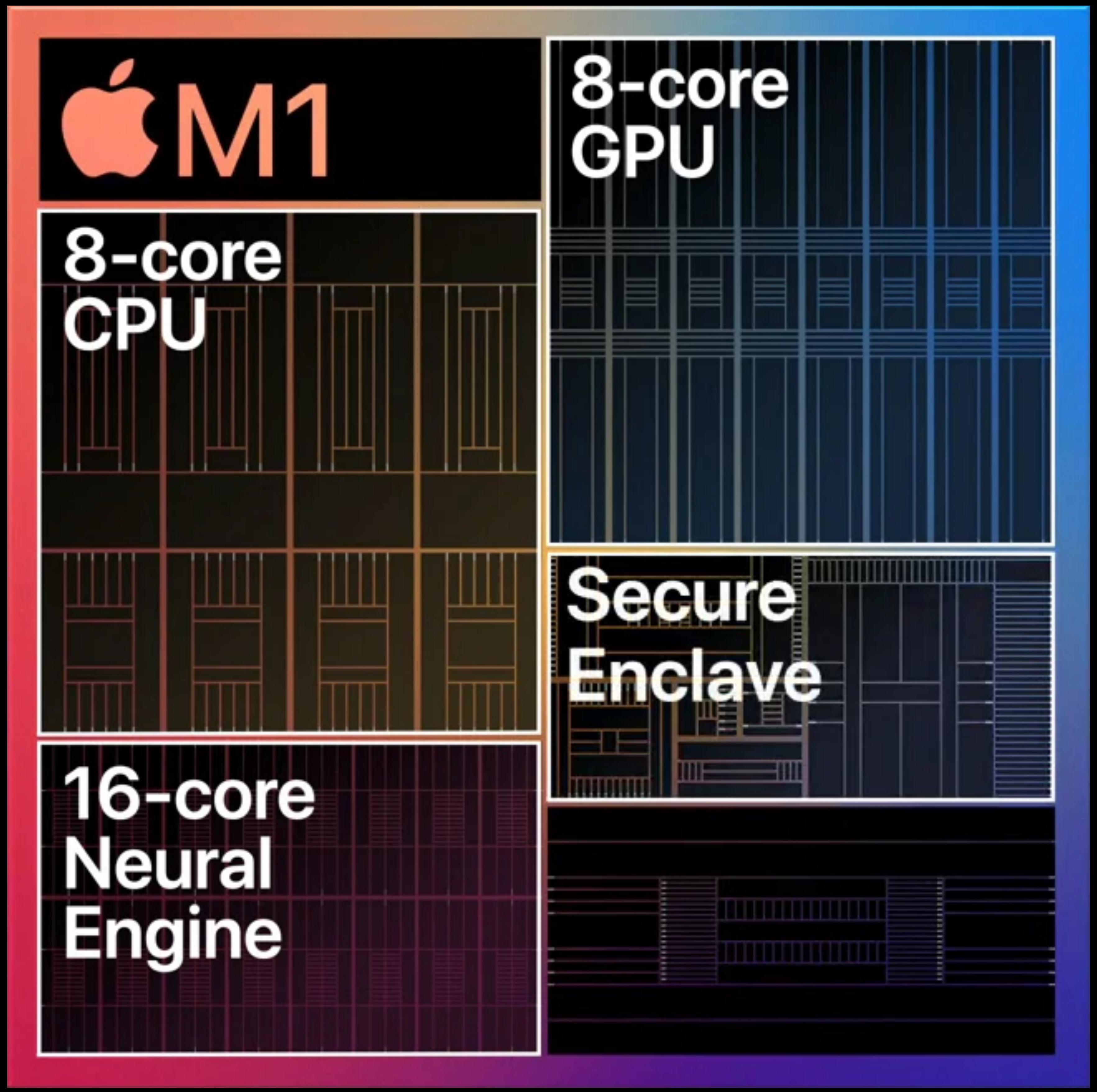
Shaders



Up to **8** cores
128 execution units
24 ALU's
Up to 24,576 concurrent threads
2.6 teraflops
82 gigatexels/second
41 gigapixels/second

OpenGL/OpenCL, Vulkan, WebGPU, Metal

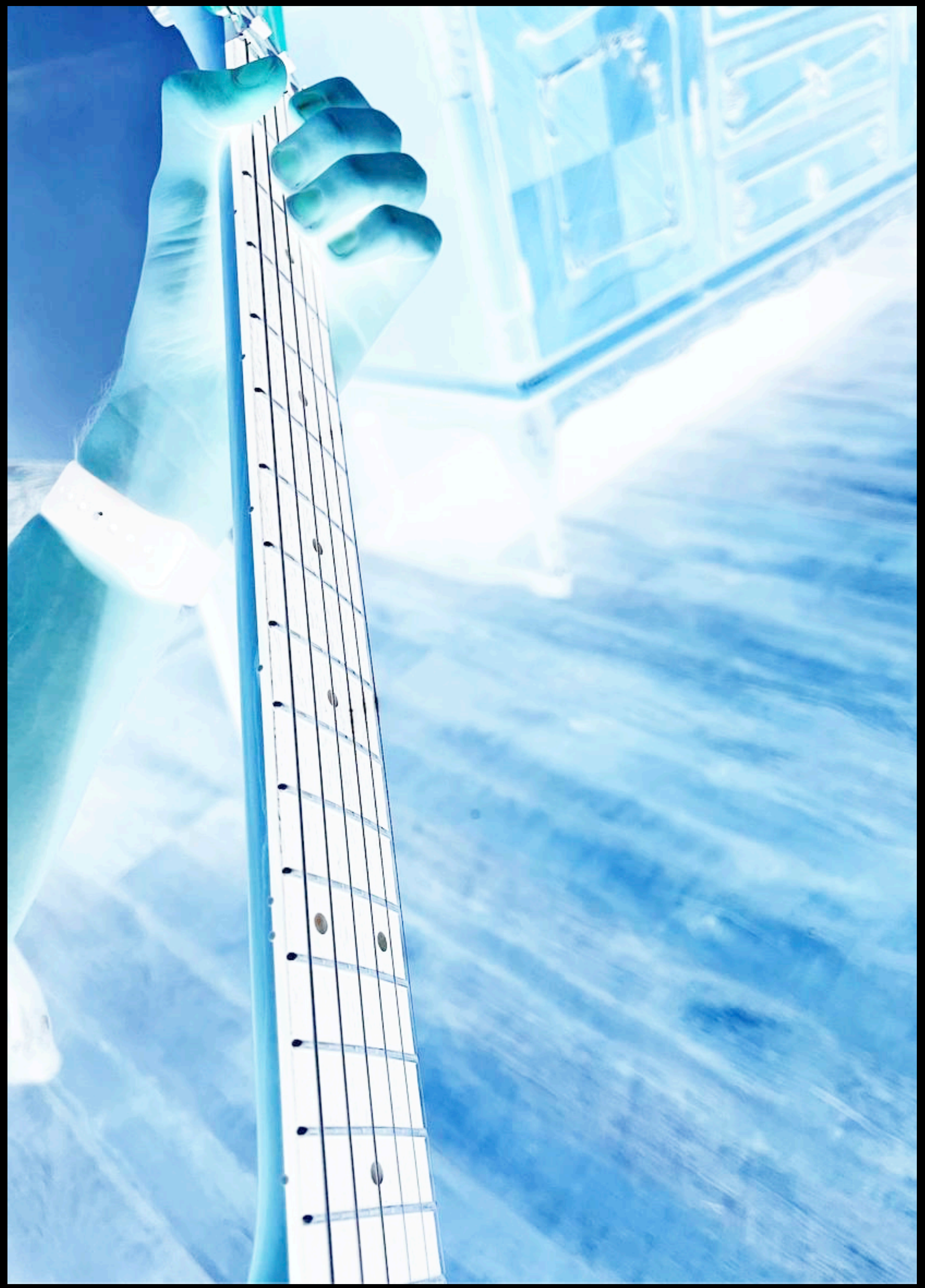
Point operations

Brightness

```
const float4 inputPixelValue = inputTexture.read(pixel_index);
outputTexture.write(inputPixelValue * brightnessFactor, pixel_index);
```

Point operations

Color inversion



```
const float4 pixel_value = inputTexture.read(pixel_index);
outputTexture.write(1 - pixel_value, pixel_index);
```

Point operations
LUT's

Point operations

Grayscale

1920 * 1080 = 2 073 600 pixels
8 bit
3 channels
6.2 Mbyte/frame
744 Mbyte/sec

Point operations

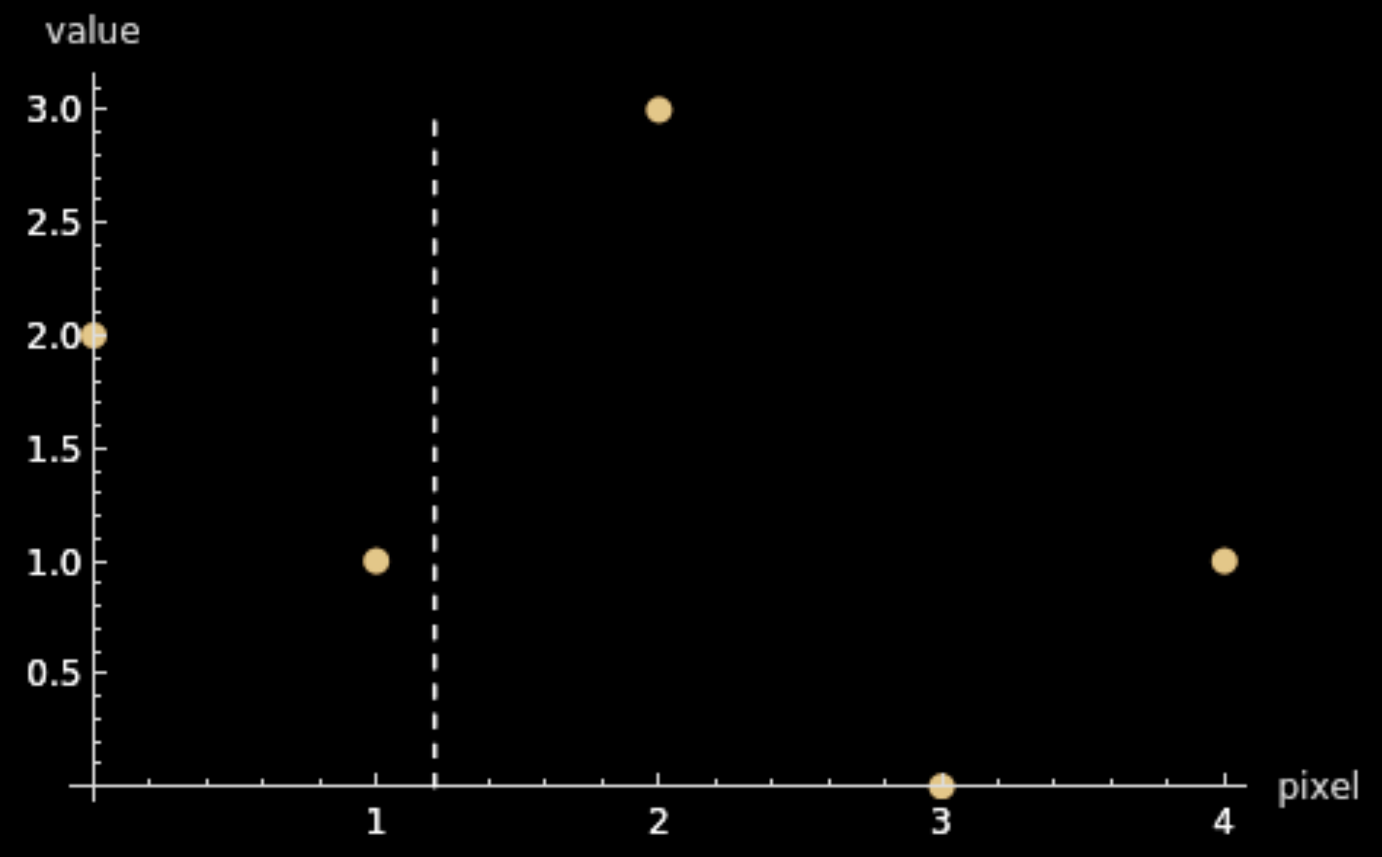
Grayscale

```
const float3 inColor = inputTexture.read(pixel_index).rgb;
float grayColor = dot(inColor, weights);

outputTexture.write(float4(grayColor, grayColor, grayColor, 1), pixel_index);
```

```
public static var rec709 = SIMD3(x: 0.2126, y: 0.7152, z: 0.0722)
public static var rec601 = SIMD3(x: 0.299, y: 0.587, z: 0.114)
public static var average = SIMD3(x: 1/3, y: 1/3, z: 1/3)
```

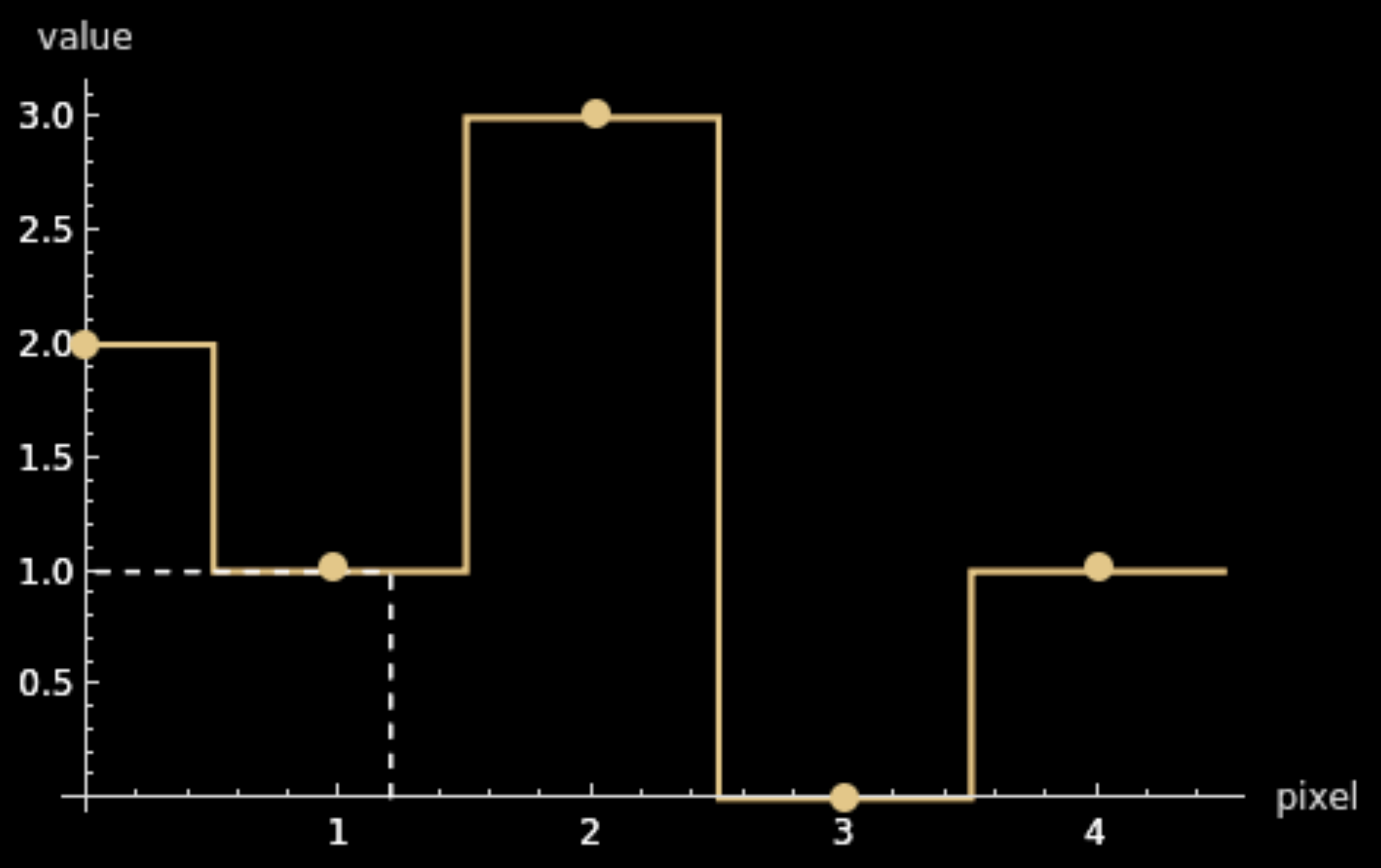
Resize



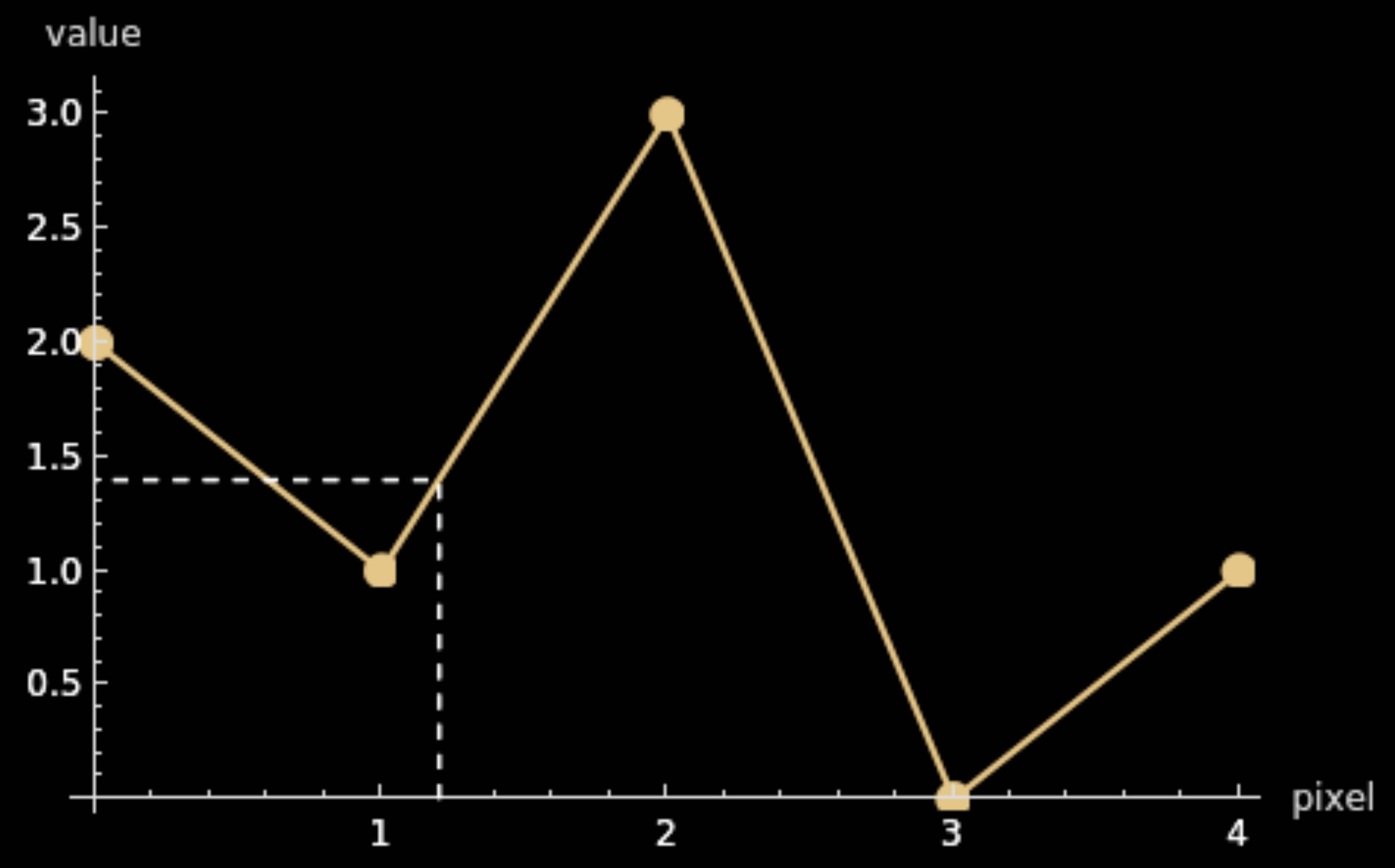
k = 1.2
pixel_i = 1
value = f(pixel_i * k)

Resize

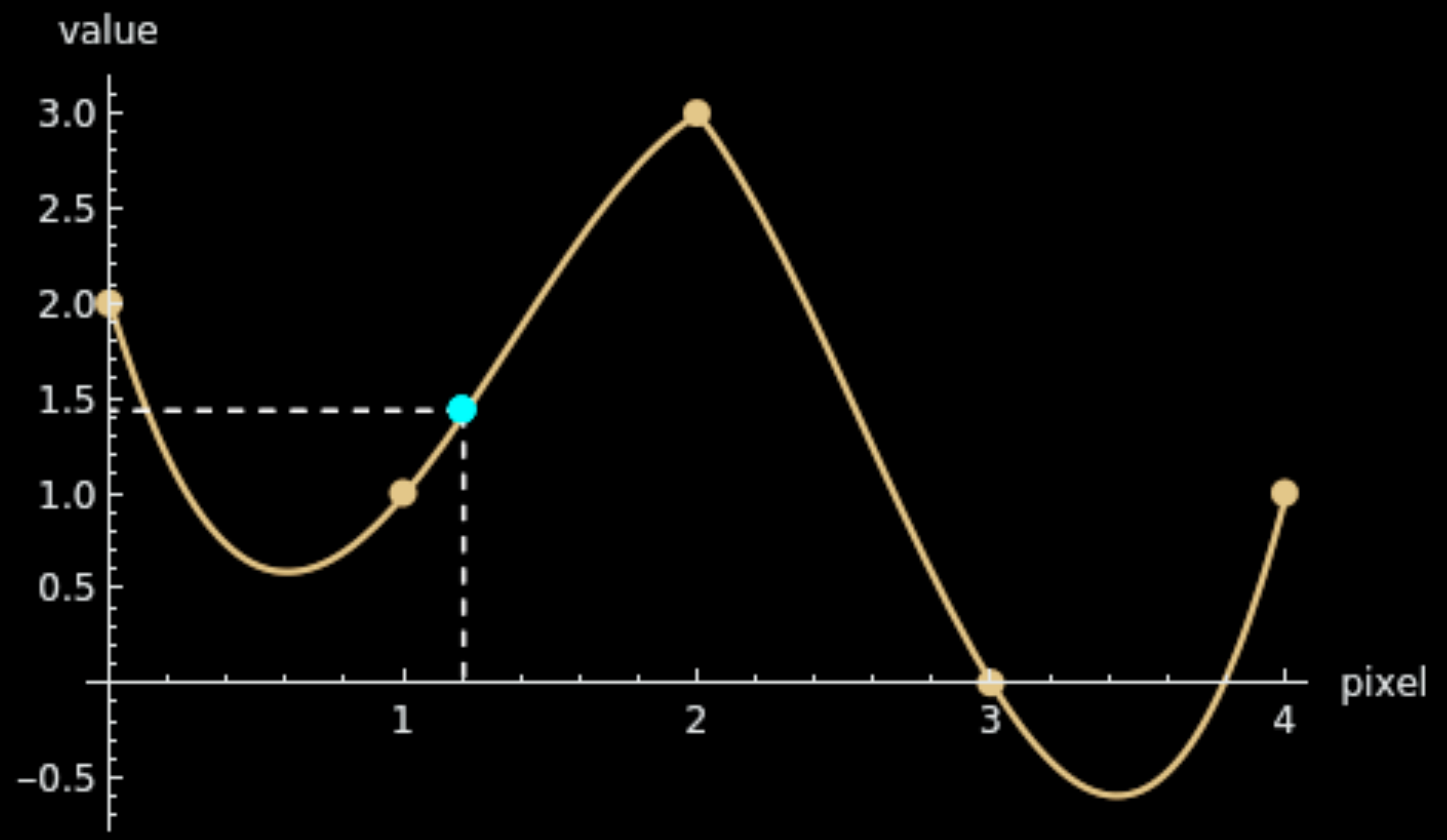
1) Nearest Neighbour



2) Bilinear Interpolation



3) Bicubic Interpolation



$k = 1.2$, value = $f(k * x_i)$

Convolutions

1) Convolutional neural networks

2) Analysis and transformation of signals (sound, radio, image), spectrograms.

3) Differentials, polynomials

4) Impulse responses

5) …

Convolutions

a = [1, 2, 3, 4]   b = [1, 2, 3, 4]

1) a + b = [2, 4, 6, 8]

2) a • b = [1, 4, 9, 16]

3) a x b = 1*1 + 2*2 + 3*3 + 4*4 = 31

Convolutions

$$(a * b)_n = \sum_{i+j=n} a_i \cdot b_j$$

Convolutions

$$(a * b)_n = \sum_{i+j=n} a_i \cdot b_j$$

$a = [\ 1, 2, 3, 4\ ]$

$b = [\ 1, 2, 3, 4\ ]$

$a * b = [\ \boxed{1}, 4, 10, 20, 25, 24, 16\ ]$

Convolutions

$$(a * b)_n = \sum_{i+j=n} a_i \cdot b_j$$

$$(a * b)_0 = a_0 \cdot b_0$$

$$(a * b)_1 = a_0 \cdot b_1 + a_1 \cdot b_0$$

$$(a * b)_2 = a_0 \cdot b_2 + a_1 \cdot b_1 + a_2 \cdot b_0$$

$$(a * b)_3 = a_0 \cdot b_3 + a_1 \cdot b_2 + a_2 \cdot b_1 + a_3 \cdot b_0$$

$$(a * b)_4 = a_0 \cdot b_4 + a_1 \cdot b_3 + a_2 \cdot b_2 + a_3 \cdot b_1 + a_4 \cdot b_0$$

$$(a * b)_5 = a_0 \cdot b_5 + a_1 \cdot b_4 + a_2 \cdot b_3 + a_3 \cdot b_2 + a_4 \cdot b_1 + a_5 \cdot b_0$$

a = [1, 2, 3, 4]

b = [1, 2, 3, 4]

a * b = [$\boxed{1}$, 4, 10, 20, 25, 24, 16]

Convolutions

$$(a * b)_n = \sum_{i+j=n} a_i \cdot b_j$$

$$j = n - i$$

$$(a * b)_n = \sum a_i \cdot b_{n-i}$$

$$[\,\boxed{1}, 2, 3, 4\,]$$

$$[\,4, 3, 2, \boxed{1}\,]$$

$$a * b = [\,\boxed{1}, 4, 10, 20, 25, 24, 16\,]$$

Convolutions

$$(a * b)_n = \sum_{i+j=n} a_i \cdot b_j = \sum a_i \cdot b_{n-i}$$

	a_0	a_2	a_2
b_0			
b_1			
b_2			
b_3			

Convolutions

$$(a * b)_n = \sum_{i+j=n} a_i \cdot b_j = \sum a_i \cdot b_{n-i}$$

	a_0	a_2	a_2
b_0	$a_0 \cdot b_0$	$a_1 \cdot b_0$	$a_2 \cdot b_0$
b_1	$a_0 \cdot b_1$	$a_1 \cdot b_1$	$a_2 \cdot b_1$
b_2	$a_0 \cdot b_2$	$a_1 \cdot b_2$	$a_2 \cdot b_2$
b_3	$a_0 \cdot b_3$	$a_1 \cdot b_3$	$a_2 \cdot b_3$

$$(a * b)_0 = a_0 \cdot b_0$$

$$(a * b)_1 = a_0 \cdot b_1 + a_1 \cdot b_0$$

$$(a * b)_2 = a_0 \cdot b_2 + a_1 \cdot b_1 + a_2 \cdot b_0$$

$$(a * b)_3 = a_0 \cdot b_3 + a_1 \cdot b_2 + a_2 \cdot b_1$$

$$(a * b)_4 = a_1 \cdot b_3 + a_2 \cdot b_2$$

$$(a * b)_5 = a_0 \cdot b_5$$

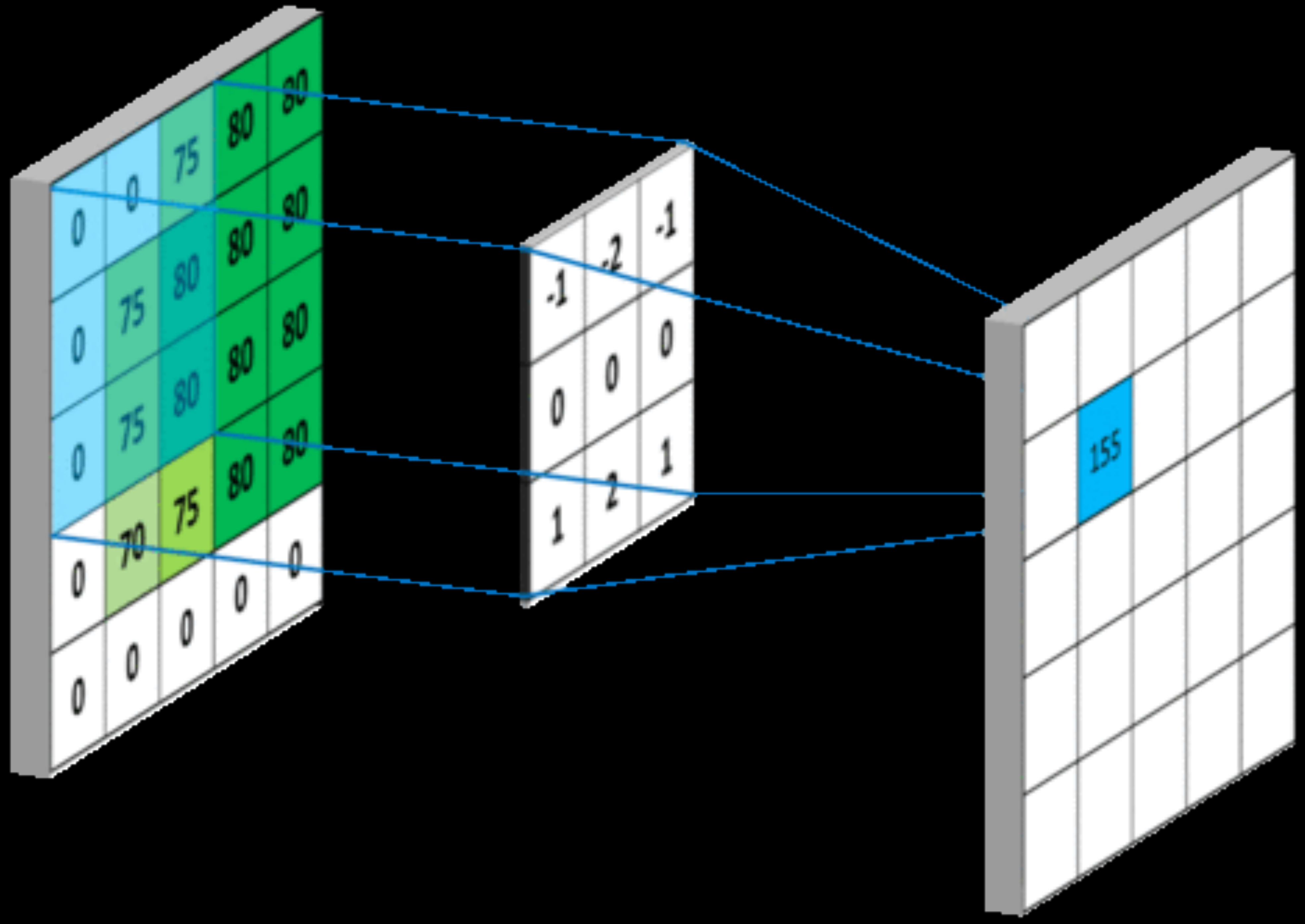
$N = a_{max} + b_{max} - 1$

Convolutions, Moving average

Kernel: [1/3, 1/3, 1/3]

Two-dimensional convolutions

$$(a * b)_{[i,\, j]} = \sum_{m=1}^{M} \sum_{n=1}^{N} a_{[m,\, n]} \cdot b_{[i - m,\, j - n]}$$



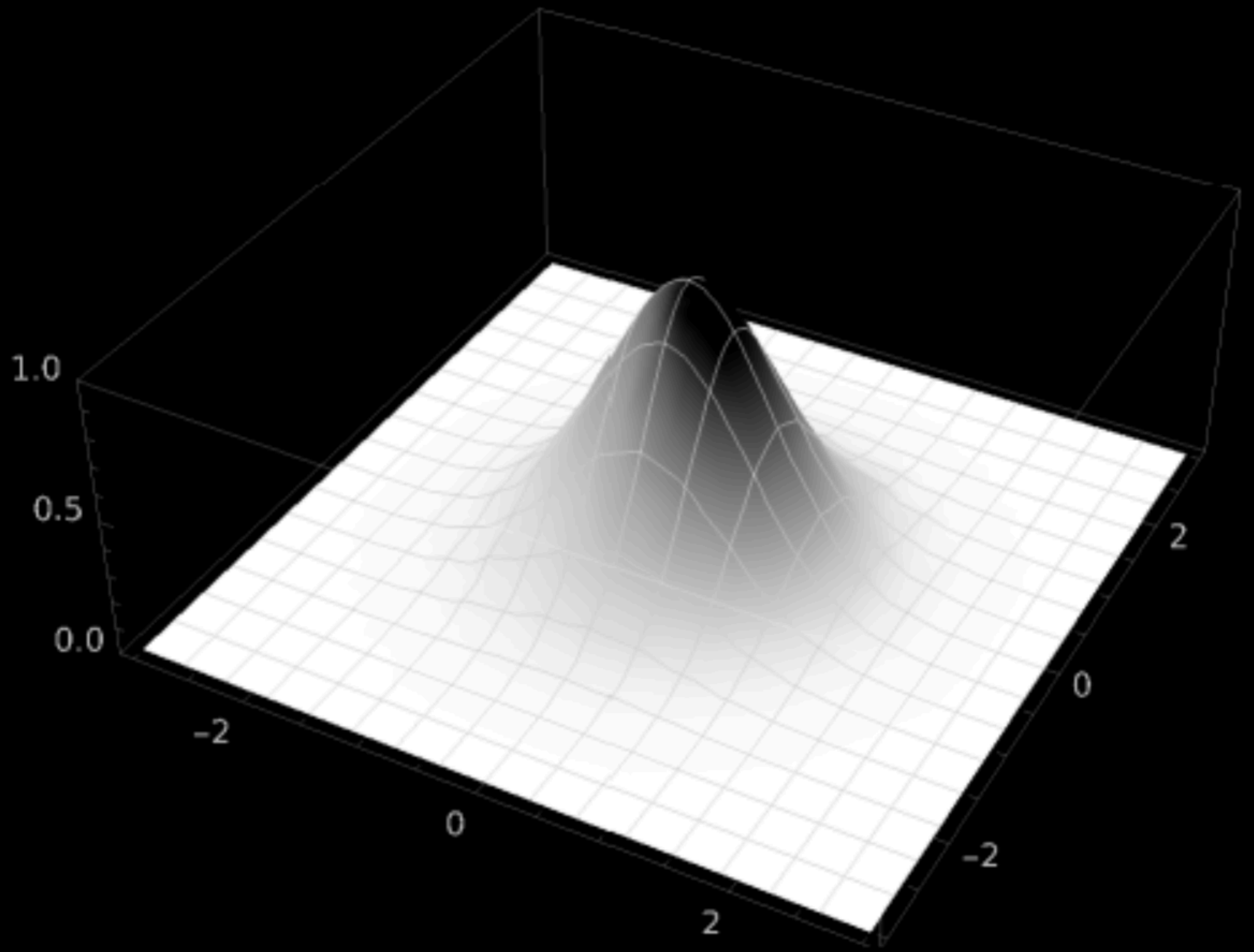
Blur, average sum

- Equal weight
- Erased borders
- “Blocked” image

1/9	1/9	1/9
1/9	1/9	1/9
1/9	1/9	1/9

Blur, Gaussian

$$G(\sigma)[i,j] = \frac{1}{2\pi\sigma^2} * e^{-\frac{(i^2+j^2)}{2\sigma^2}}$$



$\frac{1}{273}$

1	4	7	4	1
4	16	26	16	4
7	26	41	26	7
4	16	26	16	4
1	4	7	4	1

Feature extraction

How our brain processes images

$I = f(x, y)$, I — brightness at point x,y

$$\frac{\partial f}{\partial x} = \lim_{\Delta x \to 0} \frac{f(x + \Delta x, y) - f(x, y)}{\Delta x}$$

$$\frac{\partial f}{\partial y} = \lim_{\Delta y \to 0} \frac{f(x, y + \Delta y) - f(x, y)}{\Delta y}$$

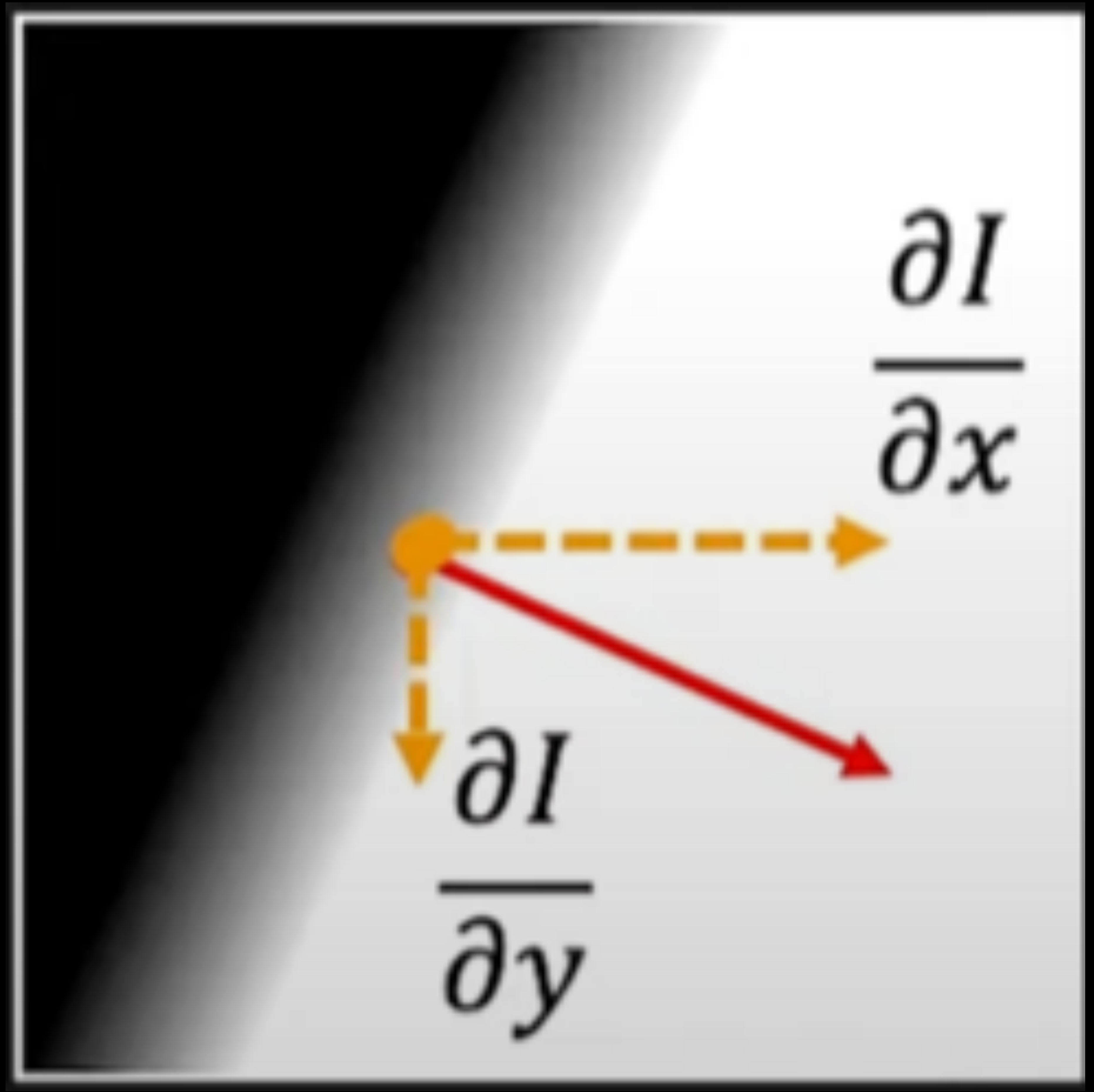
$$\left[\frac{\partial f}{\partial x}; \frac{\partial f}{\partial y}\right]$$

Feature extraction

$$v = \left[\frac{\partial f}{\partial x}; \frac{\partial f}{\partial y}\right]$$

$$|\,v\,| = \sqrt{\left(\frac{\partial f}{\partial x}\right)^2 + \left(\frac{\partial f}{\partial y}\right)^2}$$

$$\theta = \text{cotan}\left(\frac{\partial f}{\partial x} / \frac{\partial f}{\partial y}\right)$$



Feature extraction, first derivative

$$\frac{\partial f}{\partial x} \approx f(x+1, y) - f(x, y) \qquad \frac{\partial f}{\partial y} \approx f(x, y+1) - f(x, y)$$

	Roberts	Prewitt	Sobel
∂ƒ/∂x	$\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$	$\begin{bmatrix} -1 & 0 & 1 \\ -1 & 0 & 1 \\ -1 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$
∂ƒ/∂y	$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$	$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ -1 & -1 & -1 \end{bmatrix}$	$\begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{bmatrix}$

Feature extraction, first derivative

Sobel

$\partial f/\partial x$

$\partial f/\partial y$

$\partial f/\partial x + \partial f/\partial y$

Feature extraction, thresholding

Sobel

```
var isEdge = Magnitude >= threshold
```

Feature extraction, second derivative

$$\frac{\partial f}{\partial x} = I_x{}' \approx f(x+1, y) - f(x, y)$$

$$\frac{\partial^2 f}{\partial^2 x} = I_x{}'' \approx I_x{}' - I_{x-1}{}'$$

Feature extraction, second derivative

$$\frac{\partial f}{\partial x} = I_x' \approx f(x+1, y) - f(x, y)$$

$$\frac{\partial^2 f}{\partial^2 x} = I_x'' \approx I_x' - I_{x-1}' \approx f(x+1, y) - f(x, y) - f(x, y) + f(x-1, y) = f(x+1, y) - 2f(x, y) + f(x-1, y)$$

Feature extraction, second derivative

$$\frac{\partial f}{\partial x} = I_x{'} \approx f(x+1, y) - f(x, y)$$

$$\frac{\partial^2 f}{\partial^2 x} \approx f(x+1, y) - 2f(x, y) + f(x-1, y)$$

$$\frac{\partial^2 f}{\partial^2 y} \approx f(x, y+1) - 2f(x, y) + f(x, y - 1)$$

Feature extraction, second derivative

$$\frac{\partial f}{\partial x} = I_x' \approx f(x+1, y) - f(x, y)$$

$$\frac{\partial^2 f}{\partial^2 x} \approx f(x+1, y) - 2f(x, y) + f(x-1, y)$$

$$\frac{\partial^2 f}{\partial^2 y} \approx f(x, y+1) - 2f(x, y) + f(x, y - 1)$$

$$I'' \approx I_x'' + I_y'' = f(x+1, y) + f(x-1, y) + f(x, y+1) + f(x, y - 1) - 4f(x, y)$$

0	1	0
1	-4	1
0	1	0

Feature extraction, second derivative

$$\frac{\partial f}{\partial x} = I_x' \approx f(x+1, y) - f(x, y)$$

$$\frac{\partial^2 f}{\partial^2 x} \approx f(x+1, y) - 2f(x, y) + f(x-1, y)$$

$$\frac{\partial^2 f}{\partial^2 y} \approx f(x, y+1) - 2f(x, y) + f(x, y-1)$$

$$I'' \approx I_x'' + I_y'' = f(x+1, y) + f(x-1, y) + f(x, y+1) + f(x, y-1) - 4f(x, y)$$

0	1	0
1	-4	1
0	1	0

1	4	1
4	-20	4
1	4	1

Feature extraction, Laplasian

Hough transform

$y = k * x + b$

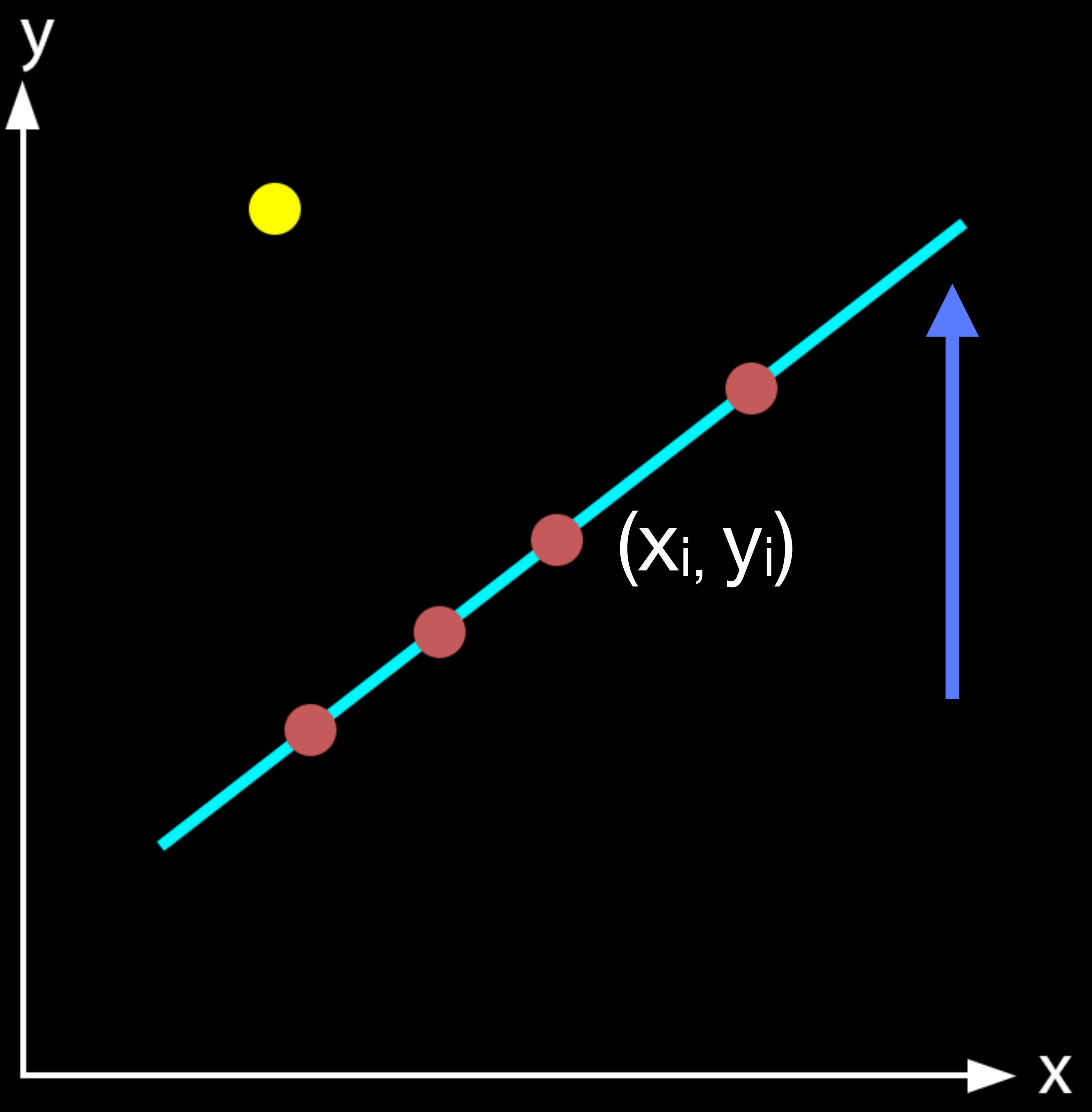
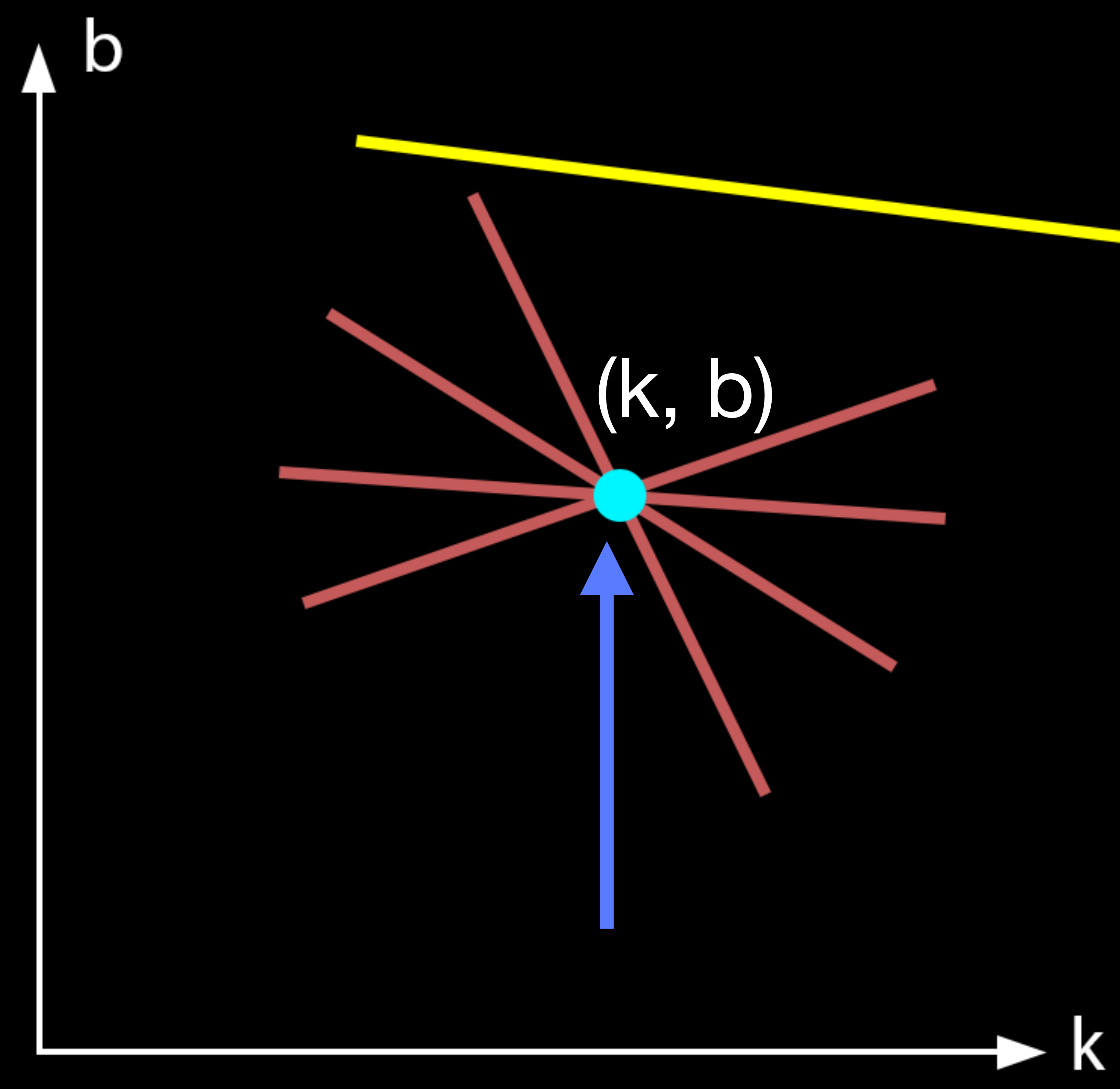


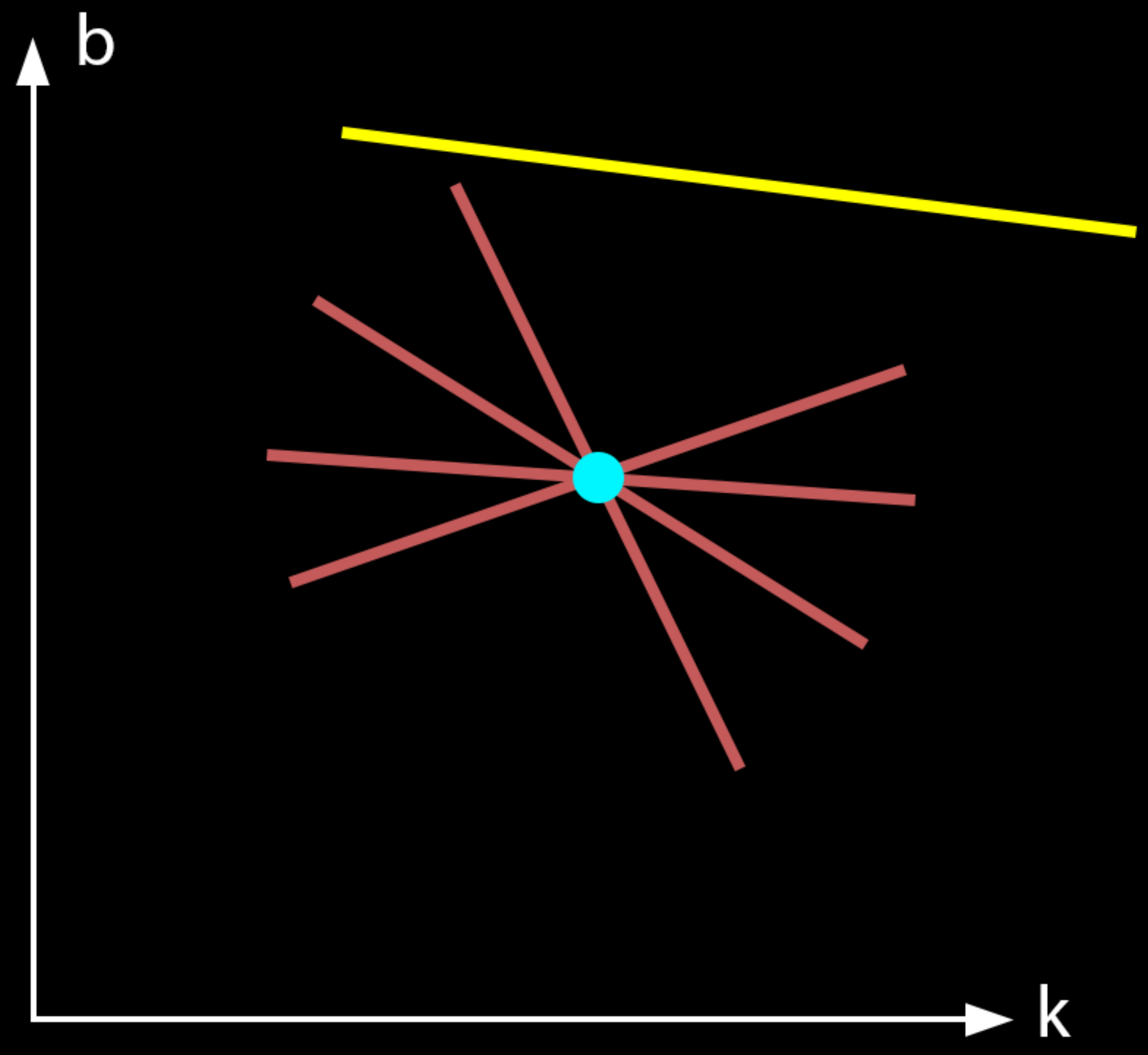
Image space

$b = -x_i * k + y_i$

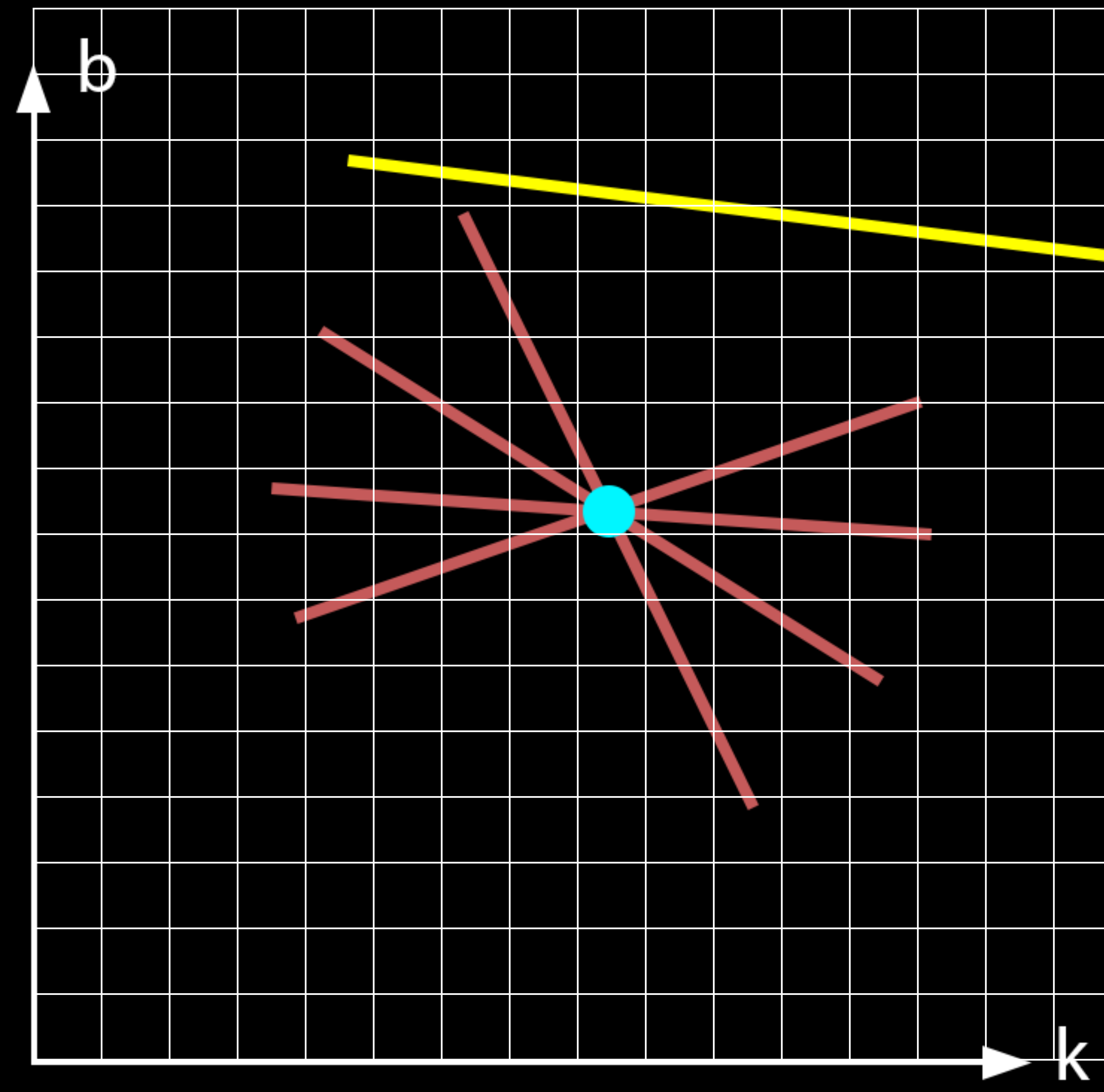


Parameter space

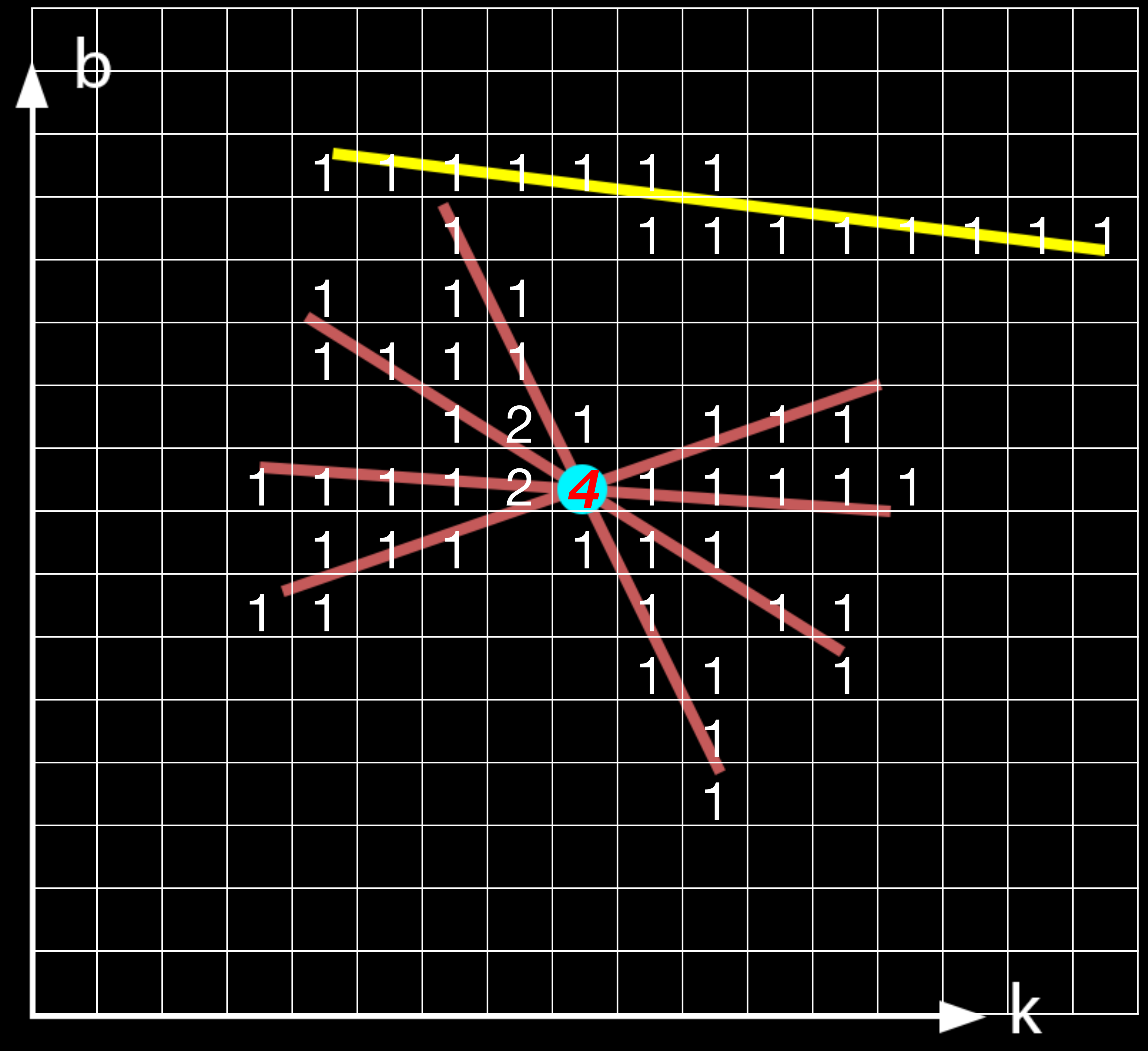
Hough transform, discretization



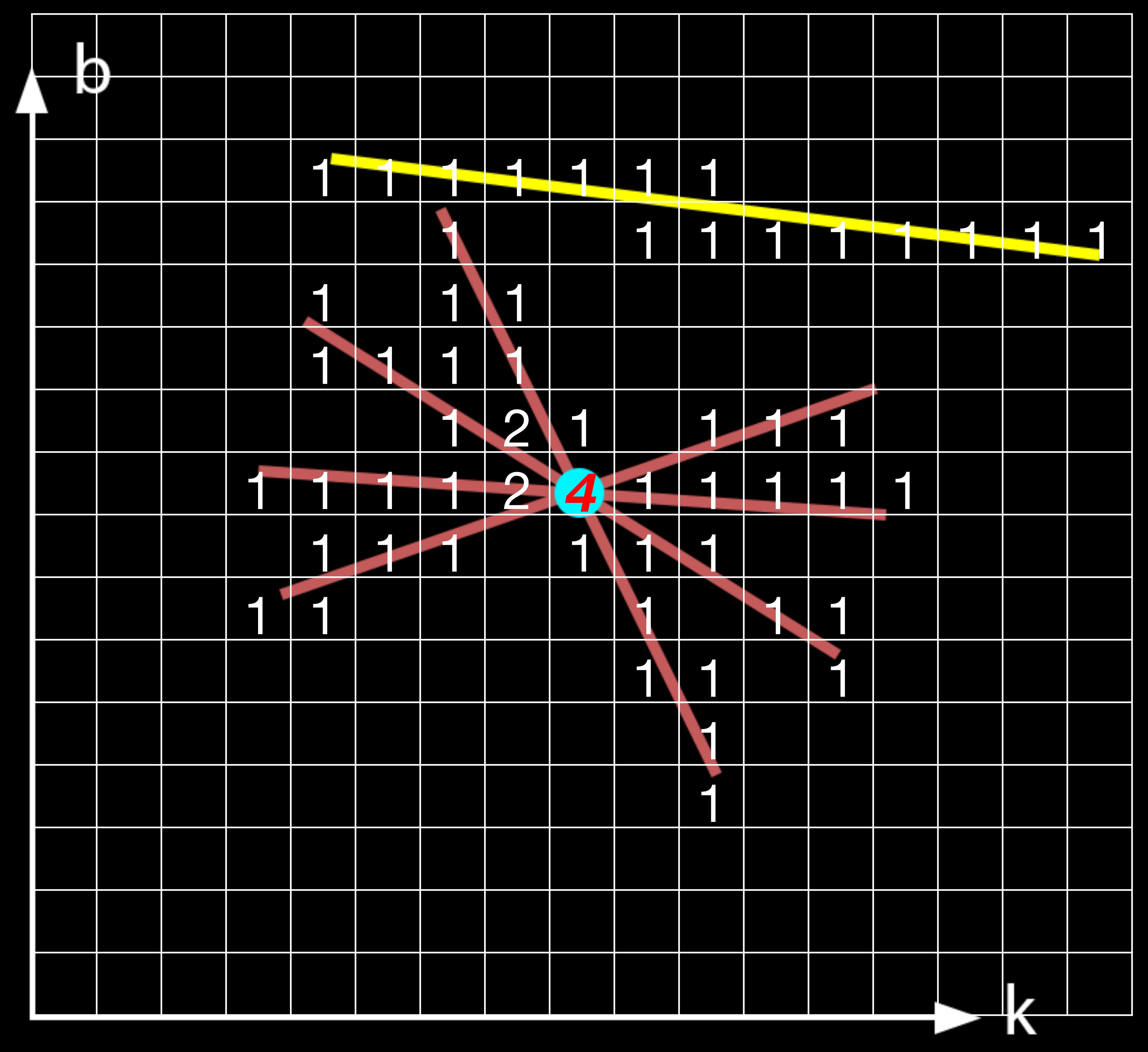
Hough transform, discretization

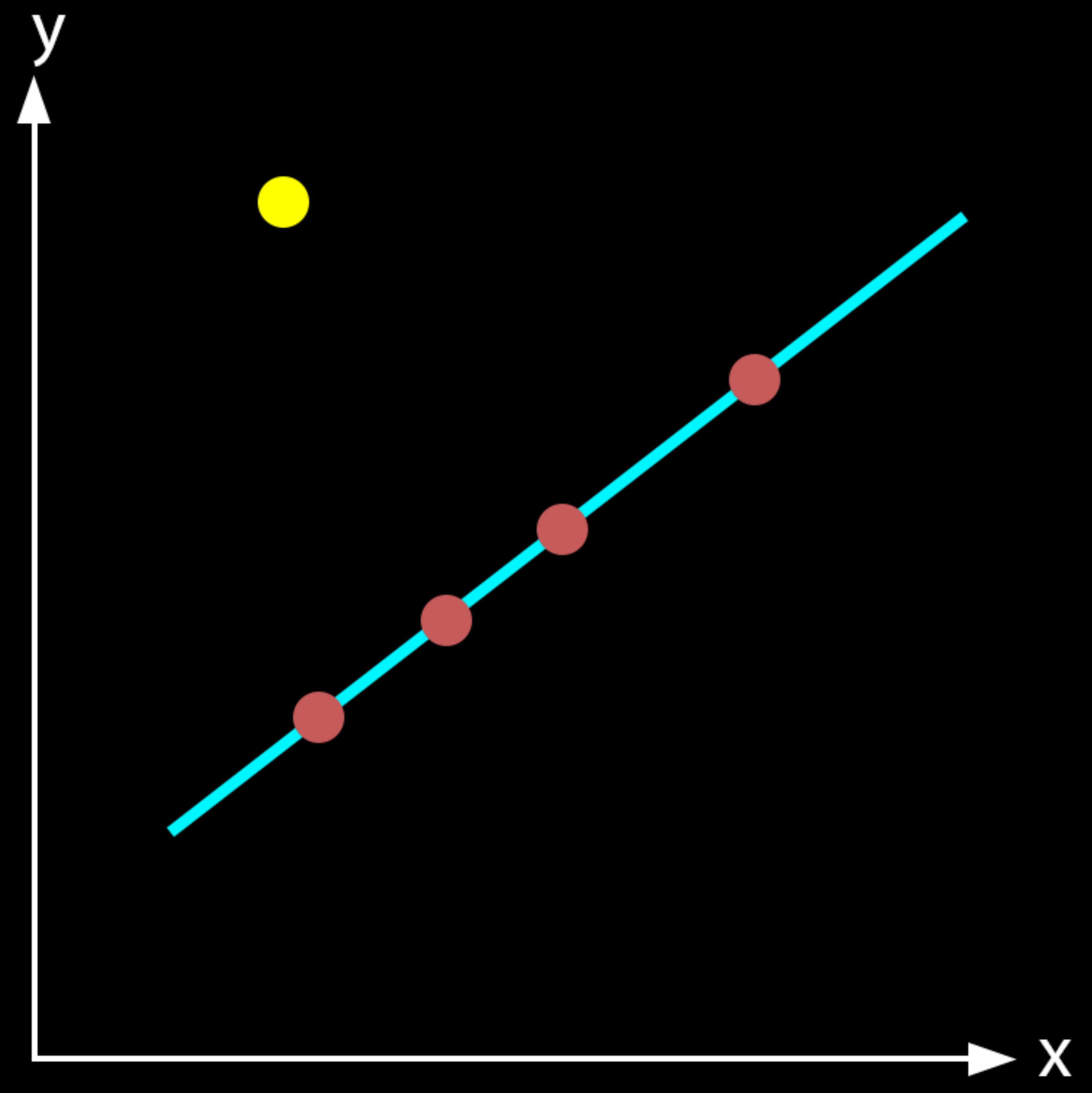


Hough transform, discretization



Hough transform, discretization

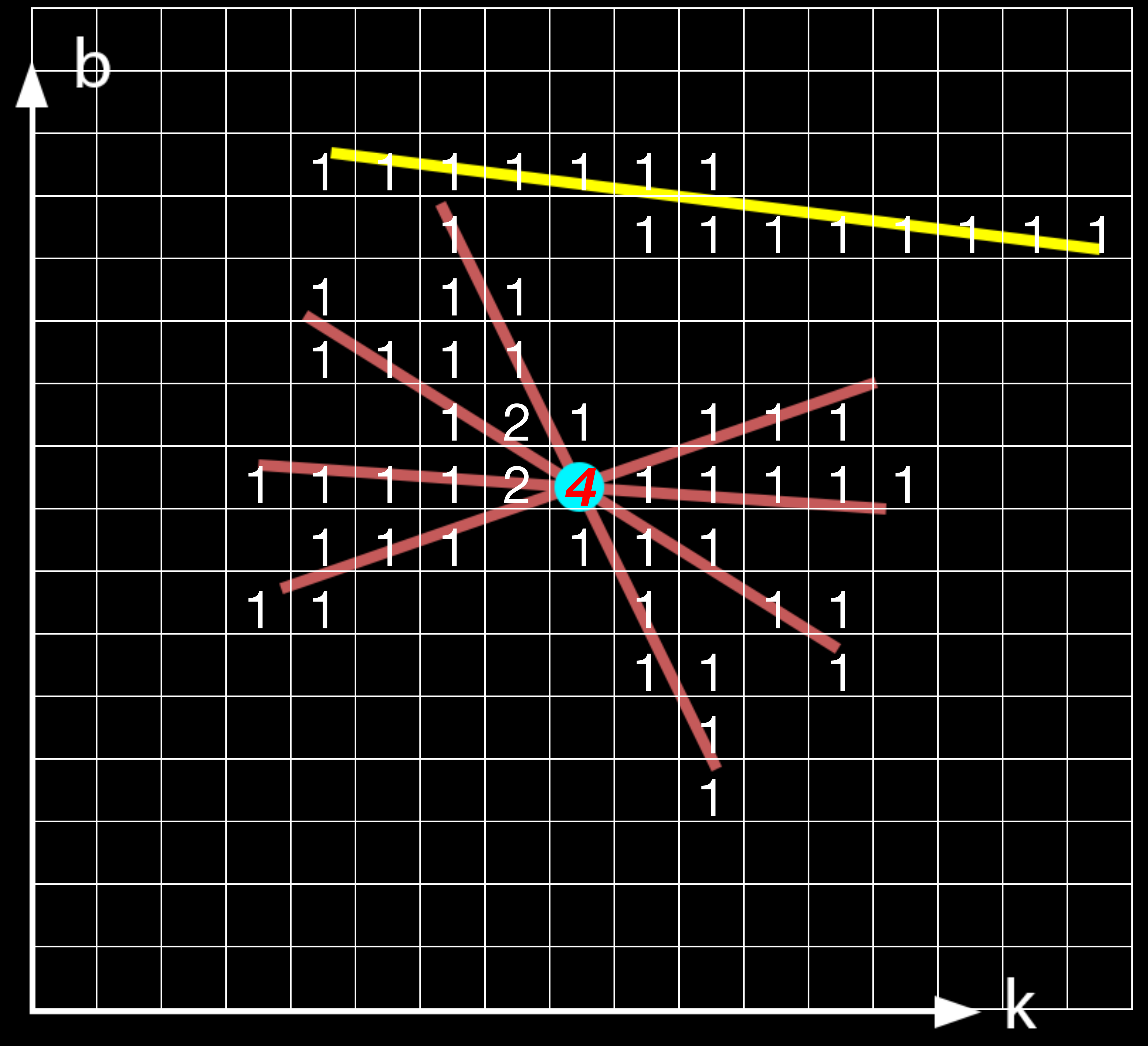




$(k, b)_{max}$ — the desired line in Image Space

$y = k_{max} * x + b_{max}$

Hough transform, discretization



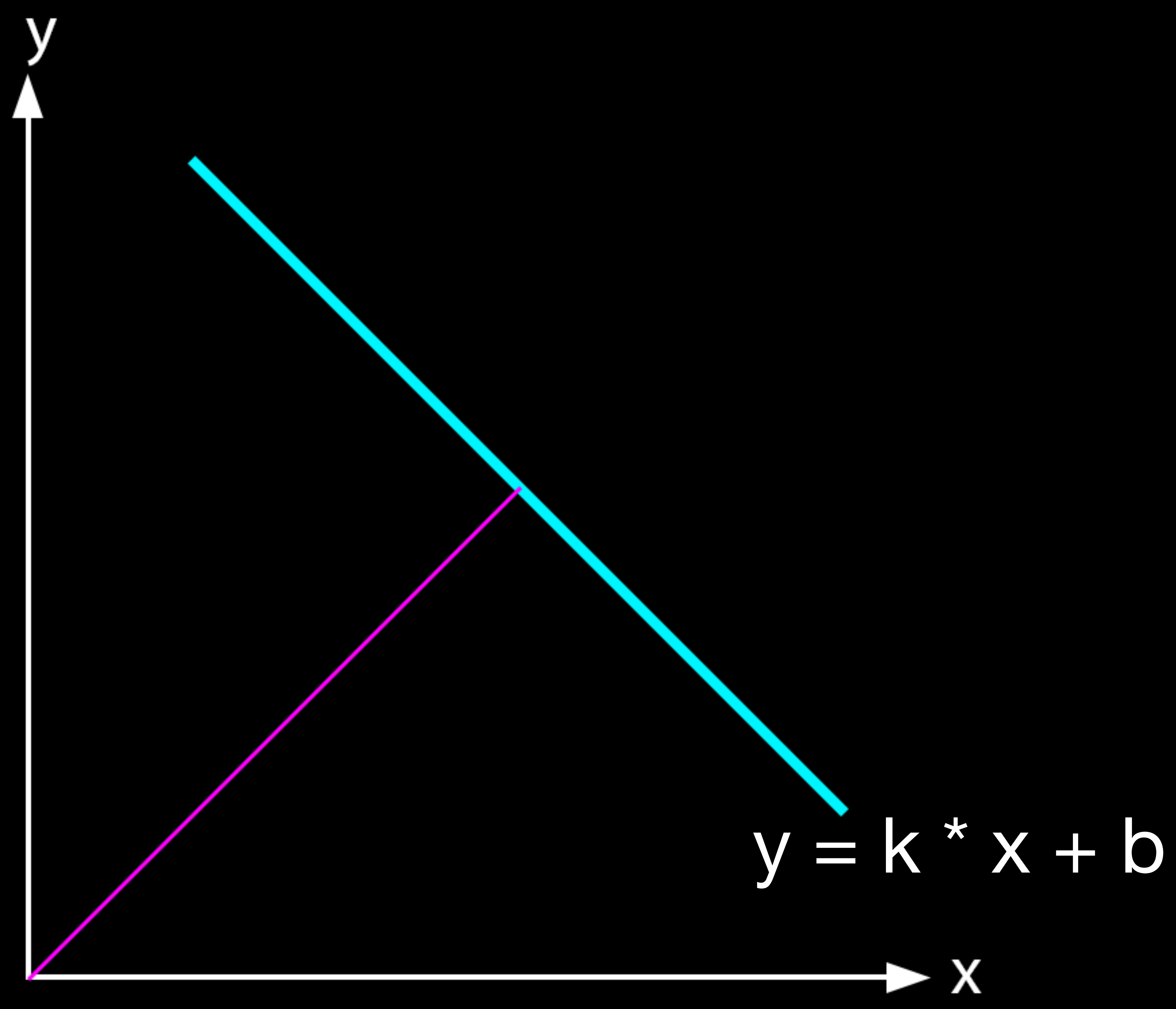
$-\infty < k,b < +\infty$

$\text{Int.min} < k,b < \text{Int.max}$

$k,b \in [2^{\wedge 0}, 2^{\wedge 10}]$

$(k, b)_{max}$ — the desired line in Image Space

Hough transform, parameterization



$$y * \cos(\theta) = x * \sin(\theta) + \rho$$

$$|\rho| \leq \sqrt{M^2 + N^2}$$

$$0 \leq \theta \leq \pi$$

Hough transform, result